



**BAULKHAM HILLS HIGH SCHOOL  
YEAR 11 P3 YEARLY EXAMINATION 2022  
MATHEMATICS ADVANCED**

**FINAL MARK**

**MARKING COVER SHEET**

**Name:** \_\_\_\_\_ **Teacher:** \_\_\_\_\_

|       |                 | Outcomes |     |    |     |     |     |       |
|-------|-----------------|----------|-----|----|-----|-----|-----|-------|
| Qn    | Parts           | F1       | T1  | T2 | C1  | E1  | S1  | Total |
|       | Multiple Choice | 1,2,5,9  | 4   | 10 | 7   | 6   | 3,8 |       |
|       |                 | /4       | /1  | /1 | /1  | /1  | /2  | /10   |
| 11    |                 | /2       |     |    |     |     |     | /2    |
| 12    |                 |          |     |    | /2  |     |     | /2    |
| 13    |                 |          |     | /2 |     |     |     | /2    |
| 14    |                 | /2       |     |    |     |     |     | /2    |
| 15    |                 |          |     |    | /3  |     |     | /3    |
| 16    |                 |          |     |    |     | /3  |     | /3    |
| 17    |                 |          |     |    |     | /3  |     | /3    |
| 18    |                 |          |     |    | /3  |     |     | /3    |
| 19    |                 |          | /3  |    |     |     |     | /3    |
| 20    |                 | /2       |     |    |     |     |     | /2    |
| 21    |                 |          |     | /3 |     |     |     | /3    |
| 22    |                 | /2       |     |    |     |     |     | /2    |
| 23    |                 |          |     | /3 |     |     |     | /3    |
| 24    |                 |          |     |    | /3  |     |     | /3    |
| 25    |                 |          | /3  |    |     |     |     | /3    |
| 26    |                 |          | /4  |    |     |     |     | /4    |
| 27    |                 |          |     |    |     |     | /4  | /4    |
| 28    |                 |          |     |    | /3  |     |     | /3    |
| 29    |                 |          |     |    |     | /4  |     | /4    |
| 30    |                 |          |     |    |     |     | /6  | /6    |
| Total |                 | /12      | /11 | /9 | /15 | /11 | /12 | /70   |

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2022 PRELIMINARY YEARLY EXAMINATION

# Mathematics Advanced

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## General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 30, show relevant mathematical reasoning and/or calculations

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**Total marks:**  
**70**

**Section I – 10 marks** (pages 3 – 8)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

**Section II – 60 marks** (pages 9 – 22)

- Attempt Questions 11 – 30
- Allow about 1 hour and 45 minutes for this section

## Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

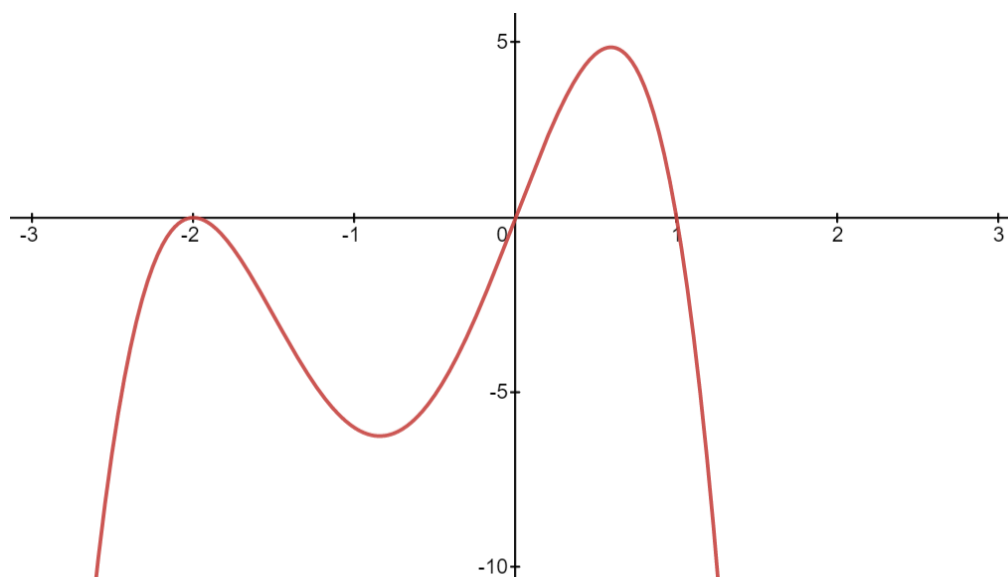
Use the multiple-choice answer page at the end of this booklet for Questions 1 – 10.

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1 What is the domain and the range of the function  $f(x) = \sqrt{4 - x^2}$  ?

- A. Domain:  $[-2, 2]$ , Range:  $[-2, 2]$
  - B. Domain:  $[-2, 2]$ , Range:  $[0, 2]$
  - C. Domain:  $[0, 2]$ , Range:  $[-4, 4]$
  - D. Domain:  $[0, 2]$ , Range:  $[0, 4]$
- 

2 Which of the following is the correct equation for the given graph?



- A.  $y = -3x(x+1)(x-2)^2$
- B.  $y = 3x(x+1)(x-2)^2$
- C.  $y = 3x(1-x)(x+2)^2$
- D.  $y = 3x(x-1)(x+2)^2$

- 3 Fifty tickets are sold in a raffle in which there are two prizes. Emily buys 5 tickets. What is the probability that Emily does not win either prize?

A.  $1 - \frac{5}{50} \times \frac{4}{49}$

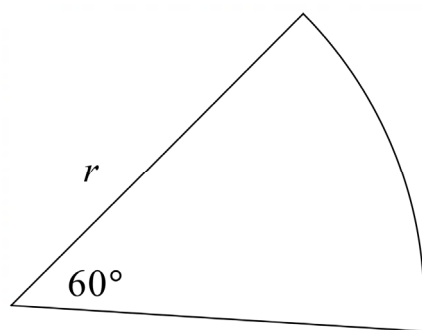
B.  $1 - \frac{5}{50} - \frac{4}{49}$

C.  $\frac{45}{50} + \frac{44}{49}$

D.  $\frac{45}{50} \times \frac{44}{49}$

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- 4 The sector below has an area of  $30\pi$  square units.



NOT TO  
SCALE

What is the value of  $r$ ?

A.  $6\sqrt{5}$

B.  $\frac{6}{\sqrt{5}}$

C.  $5\sqrt{6}$

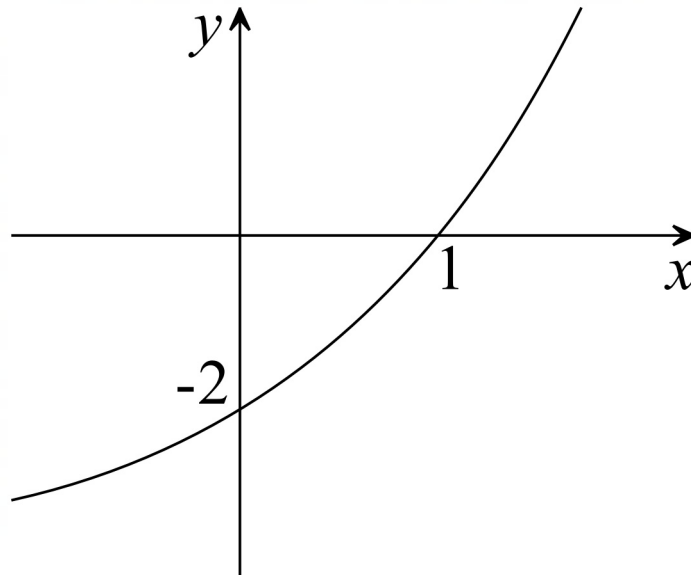
D.  $3\sqrt{2}$

- 5 Which of the following is the point of intersection of the two lines  $3x - 4y + 6 = 0$  and  $x - y - 1 = 0$ ?

A. (0,0)  
B. (-2,-3)  
C. (10,9)  
D. (11,10)

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- 6 The graph of  $y = 2^{x+a} + b$  is shown.



What are the values of  $a$  and  $b$ ?

A.  $a = 0$  and  $b = -2$   
B.  $a = 0$  and  $b = -3$   
C.  $a = 1$  and  $b = -2$   
D.  $a = 1$  and  $b = -4$

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7 What is the gradient of the normal to the curve  $y = x^2 - 4x$  at the point  $(-1, 5)$ ?

A.  $-6$

B.  $-\frac{1}{6}$

C.  $\frac{1}{6}$

D.  $6$

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8 In a class of 30 students, 8 play guitar, 12 play piano and 4 play both. What is the probability that a student selected at random either plays the guitar or plays the piano but not both?

A.  $\frac{2}{15}$

B.  $\frac{2}{5}$

C.  $\frac{8}{15}$

D.  $\frac{2}{3}$

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9 The graph of the function  $f(x) = 3x^2 - kx + 2$  is symmetrical about the line  $x = \frac{1}{2}$ .  
What is the range of the function?

A.  $y \geq \frac{1}{4}$

B.  $y \geq \frac{1}{2}$

C.  $y \geq \frac{3}{4}$

D.  $y \geq \frac{5}{4}$

**10** How many solution(s) does the equation  $\cos x(\tan x - 1) = 0$  have for  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ ?

A. 1

B. 2

C. 3

D. 4

**End of Section I**

## Section II

60 marks

Attempt Questions 11–30

Allow about 1 hour and 45 minutes for this section

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### Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing paper is available. If you use extra paper, write your name, your teacher's name on the paper and clearly indicate which question you are answering.

#### Question 11 (2 marks)

Solve  $\frac{x}{2} - (3x - 5) = 4$ . 2

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#### Question 12 (2 marks)

Differentiate  $(3x^2 + 1)^5$ . 2

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**Question 13** (2 marks)

Solve  $\sqrt{3} \tan \theta = 1$  for  $0 \leq \theta \leq 2\pi$  . **2**

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**Question 14** (2 marks)

Solve  $|x - 3| = 4$  . **2**

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**Question 15** (3 marks)

For what values of  $x$  is the function  $y = x^3 + x^2 - x + 2$  stationary? **3**

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**Question 16** (3 marks)

If  $\log_a x = 7.23$  and  $\log_a y = 2.19$ , find the value of:

(a)  $\log_a \frac{x}{y}$  **1**

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(b)  $\log_a xy^2$  **2**

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**Question 17** (3 marks)

Differentiate  $\frac{e^{3x}}{4x}$ , expressing your answer in simplest form. **3**

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**Question 18** (3 marks)

Find the equation of the tangent to the curve  $y = \sqrt{x}$  at the point where  $x = 4$ .

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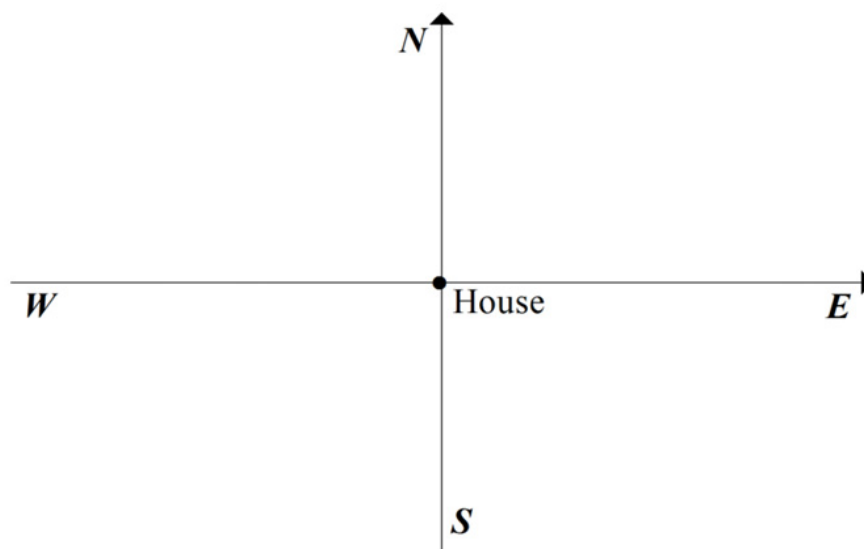
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**Question 19** (3 marks)

Asmita walks 270 metres from her house on a bearing of  $330^\circ$  to point  $B$ . She then walks in a straight line 290 metres to a point  $C$  which is due west of her house.

- (a) Clearly indicate the information above on the following diagram.

1



- (b) Calculate the distance of  $C$  from her house. Give your answer in metres, correct to 1 decimal place.

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**Question 20** (2 marks)

Write  $\frac{6}{\sqrt{5}-\sqrt{3}}$  with a rational denominator and simplify.

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**Question 21** (3 marks)

Prove  $\frac{1+\cot \theta}{\operatorname{cosec} \theta} - \frac{\sec \theta}{\tan \theta + \cot \theta} = \cos \theta$ .

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**Question 22** (2 marks)

Solve for  $x$ ,  $3^{1-x} = \frac{1}{\sqrt{27}}$ .

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**Question 23** (3 marks)

Solve  $10 \sin^2 \theta + \cos \theta - 7 = 0$ , where  $0 \leq \theta \leq 2\pi$ .

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**Question 24** (3 marks)

Differentiate, from first principles,  $f(x) = 3x^2 - 5$ .

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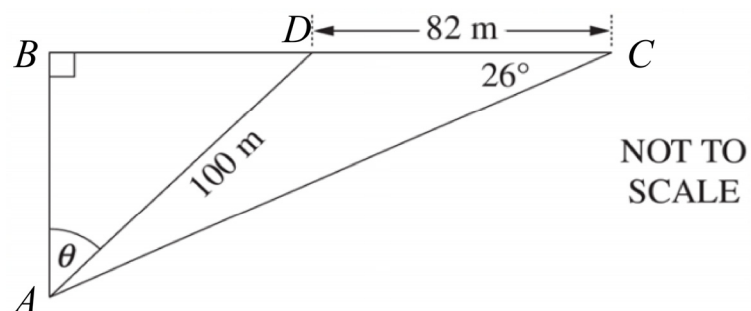
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**Question 25** (3 marks)

Calculate the value of  $\theta$  to the nearest minute.

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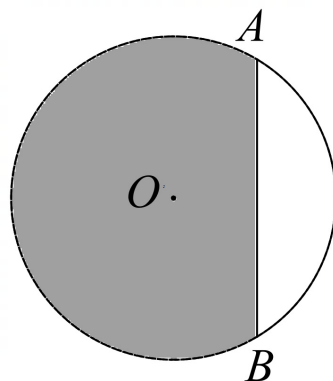
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**Question 26** (4 marks)

A garden bed is in the shape of a circle with a minor segment removed. The circle has centre,  $O$ , and a radius of 5 metres. The length of the straight edge  $AB$  is  $5\sqrt{3}$  metres.



Find the area of the garden bed, leaving your answer as an exact value.

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**Question 27** (4 marks)

James has three bags of lollies. In Bag 1, there are three mints and three toffees, in Bag 2 there are three mints and two toffees, and in Bag 3, there are two mints and one toffee.

James selects a bag at random, and then selects a lolly at random.

- (a) Find the probability he chooses a mint from Bag 1. **1**

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- (b) Find the probability he chooses a mint. **2**

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- (c) Find the probability that James chose Bag 1, given that he selects a mint. **1**

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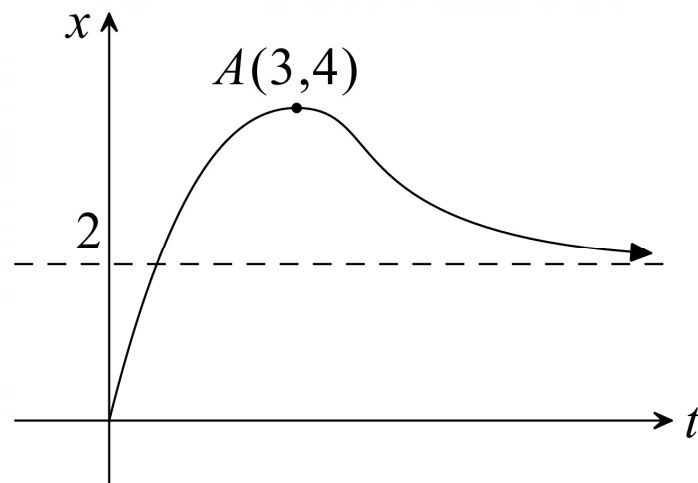
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**Question 28** (3 marks)

The graph below shows the displacement of a particle over time.



Sketch a possible graph of the particle's velocity,  $\frac{dx}{dt}$ , on the axes above.

3

**Question 29** (4 marks)

A bacteria culture of  $N$  bacteria after  $t$  minutes is increasing exponentially and can be modelled using such that  $N = Ae^{kt}$ . Initially, at 9 am, there was 100 bacteria and after 2 minutes there were 400 bacteria.

Calculate, to the nearest minute, the rate at which the bacteria are increasing at 9:10 am. **4**

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**Question 30** (6 marks)

A biased die with six faces is rolled. The discrete random variable  $X$  represents the score on the uppermost face. The probability distribution of  $X$  is shown in the table below.

|          |     |     |     |     |     |     |
|----------|-----|-----|-----|-----|-----|-----|
| $x$      | 1   | 2   | 3   | 4   | 5   | 6   |
| $P(X=x)$ | $a$ | $a$ | $a$ | $b$ | $b$ | 0.3 |

- (a) Explain why  $3a + 2b = 0.7$ . **1**

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- (b) Given that  $E(X) = 4.2$ , find the values of  $a$  and  $b$ . **3**

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**Question 30 continues on page 21**

Question 30 (continued)

(c) Find the variance of the distribution.

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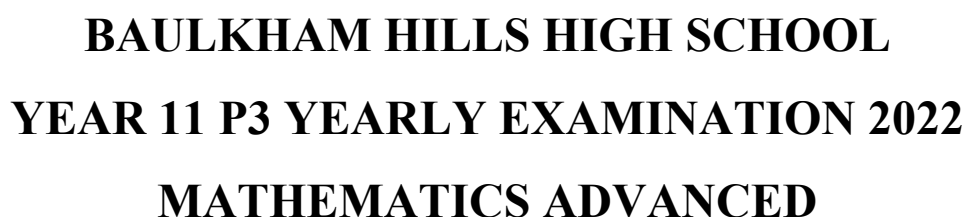
**End of Paper**

### ADDITIONAL WORKING OUT SPACE:

If you are continuing a question on this page, write “Continued on extra page” at the end of the allocated space for the question you are continuing. Write the question number clearly before continuing/starting a question on this page.

[illegible]





**Teacher:**

## Section I – Multiple Choice

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# YEAR 11 YEARLY EXAMINATION 2022 MATHEMATICS ADVANCED MARKING GUIDELINES

## Section I

### Multiple-choice Answer Key

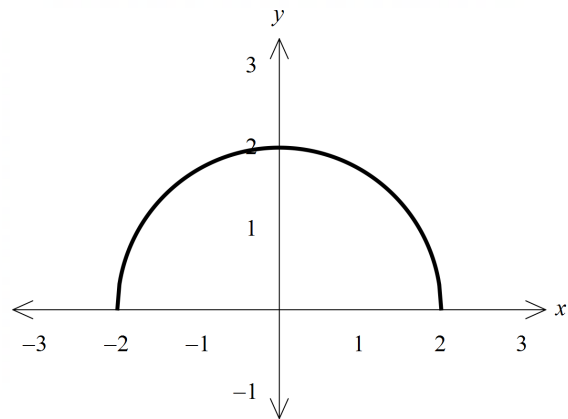
| Question | Answer |
|----------|--------|
| 1        | B      |
| 2        | C      |
| 3        | D      |
| 4        | A      |
| 5        | C      |

| Question | Answer |
|----------|--------|
| 6        | D      |
| 7        | C      |
| 8        | B      |
| 9        | D      |
| 10       | A      |

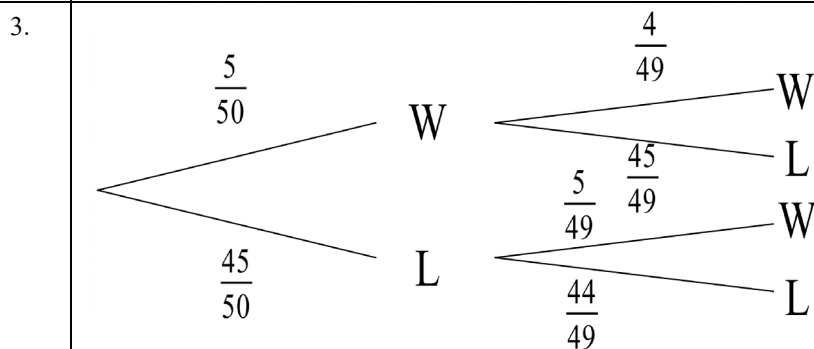
### Questions 1 – 10

#### Sample solution

1. Semi-circle, radius 2 and  $y \geq 0$ .  
Therefore B.



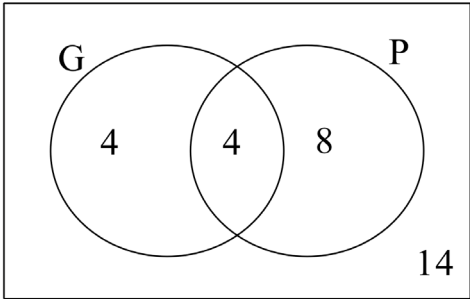
2. Zeroes at -2, 0 and 1, therefore C or D.  
Leading coefficient is negative, as  $x \rightarrow \infty, y \rightarrow -\infty$ .  
Therefore C.



$$P(LL) = \frac{45}{50} \times \frac{44}{49}$$

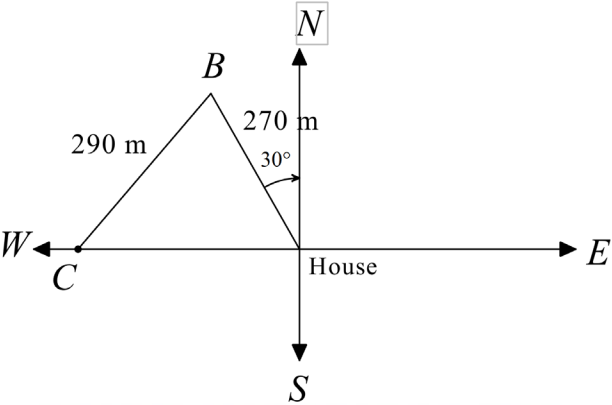
Therefore D.

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. | $A = \frac{1}{2}r^2\theta$ $30\pi = \frac{1}{2}r^2 \frac{\pi}{3}$ $180\pi = r^2\pi$ $r^2 = 180$ $r = \sqrt{180}$ $r = 6\sqrt{5}$ <p>Therefore A.</p>                                                                                                                                                                                                                                                                                                                                                                                |
| 5. | $3x - 4y + 6 = 0$ $3x - 3y - 3 = 0$ <p>Subtracting gives:</p> $-y + 9 = 0$ $y = 9$ <p>Therefore C.</p>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6. | <p>Test A:</p> $y = 2^x - 2$ <p>When <math>x = 0</math>,</p> $y = 2^0 - 2$ $y = -1 \text{ (not A)}$ <p>Test B:</p> $y = 2^x - 3$ <p>When <math>x = 0</math>, <math>y = -2</math></p> <p>When <math>x = 1</math>, <math>y = -1</math> (not B)</p> <p>Test C:</p> $y = 2^{x+1} - 2$ <p>When <math>x = 0</math>, <math>y = 2 - 2</math></p> $y = 0 \text{ (not C)}$ <p>Test D:</p> $y = 2^{x+1} - 4$ <p>When <math>x = 0</math>, <math>y = -2</math></p> <p>When <math>x = 1</math>, <math>y = 0</math> (true)</p> <p>Therefore D.</p> |

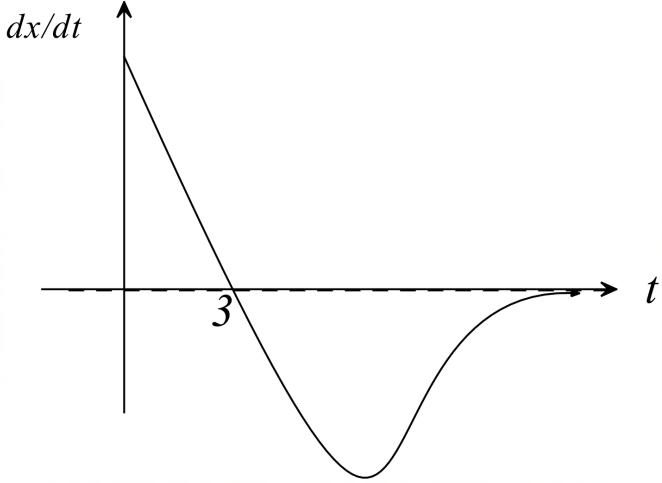
|     |                                                                                                                                                                                                                                                                                                        |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.  | $y' = 2x - 4$ <p>When <math>x = -1, y' = -6</math></p> $m_{\perp} = \frac{1}{6}$ <p>Therefore C.</p>                                                                                                                                                                                                   |
| 8.  |  <p> <math display="block">P(G \text{ or } P \text{ but not both}) = \frac{4+8}{30}</math> <math display="block">= \frac{2}{5}</math> </p> <p>Therefore B.</p>                                                        |
| 9.  | <p>Axis of symmetry:</p> $\frac{1}{2} = \frac{k}{6}$ $k = 3$ $f\left(\frac{1}{2}\right) = 3 \times \frac{1}{4} - \frac{3}{2} + 2$ $= \frac{5}{4}$ <p>Since concave up, <math>y \geq \frac{5}{4}</math>.</p> <p>Therefore D.</p>                                                                        |
| 10. | <p><math>\cos x = 0</math>      or      <math>\tan x = 1</math></p> $x = -\frac{\pi}{2}, \frac{\pi}{2} \quad \text{or} \quad x = \frac{\pi}{4}$ <p>But <math>\tan\left(-\frac{\pi}{2}\right)</math> and <math>\tan \frac{\pi}{2}</math> are undefined.</p> <p>Only 1 solution.</p> <p>Therefore A.</p> |

## Section II

| Sample solution |                                                                                                                                                                                               | Suggested marking criteria                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11.             | $\frac{x}{2} - 3x + 5 = 4$ $x - 6x + 10 = 8$ $5x = 2$ $x = \frac{2}{5}$                                                                                                                       | <ul style="list-style-type: none"> <li>• 2 – correct answer</li> <li>• 1 – correctly multiplies by 2 or correctly expands</li> </ul>                             |
| 12.             | $\frac{d}{dx}(3x^2 + 1)^5 = 5(3x^2 + 1)^4 \times 6x$ $= 30x(3x^2 + 1)^4$                                                                                                                      | <ul style="list-style-type: none"> <li>• 2 – correct answer</li> <li>• 1 – obtains <math>5(3x^2 + 1)^4</math> or <math>6x</math></li> </ul>                      |
| 13.             | $\tan \theta = \frac{1}{\sqrt{3}}$ <p>1<sup>st</sup>, 3<sup>rd</sup> quadrants</p> $\theta = \frac{\pi}{6}, \pi + \frac{\pi}{6}$ $\theta = \frac{\pi}{6}, \frac{7\pi}{6}$                     | <ul style="list-style-type: none"> <li>• 2 – correct solution</li> <li>• 1 – obtains <math>\frac{\pi}{6}</math></li> </ul>                                       |
| 14.             | $x - 3 = 4 \quad \text{or} \quad x - 3 = 4$ $x = -1 \quad \text{or} \quad x = 7$                                                                                                              | <ul style="list-style-type: none"> <li>• 2 – correct solution</li> <li>• 1 – one correct solution</li> </ul>                                                     |
| 15.             | $\frac{dy}{dx} = 3x^2 + 2x - 1$ <p>For function to be stationary, <math>\frac{dy}{dx} = 0</math></p> $3x^2 + 2x - 1 = 0$ $(3x - 1)(x + 1) = 0$ $x = -1 \quad \text{or} \quad x = \frac{1}{3}$ | <ul style="list-style-type: none"> <li>• 3 – correct solution</li> <li>• 2 – obtains <math>3x^2 + 2x - 1 = 0</math></li> <li>• 1 – correct derivative</li> </ul> |
| 16.             | <p>a) <math>\log_a \frac{x}{y} = \log_a x - \log_a y</math></p> $= 7.23 - 2.19$ $= 5.04$                                                                                                      | <ul style="list-style-type: none"> <li>• 1 – correct answer</li> </ul>                                                                                           |
|                 | <p>b) <math>\log_a xy^2 = \log_a x + \log_a y^2</math></p> $= \log_a x + 2\log_a y$ $= 7.23 + 2 \times 2.19$ $= 11.61$                                                                        | <ul style="list-style-type: none"> <li>• 2 – correct answer</li> <li>• 1 – expresses as <math>\log_a x + \log_a y^2</math></li> </ul>                            |
| 17.             | $\frac{d}{dx} \left( \frac{e^{3x}}{4x} \right) = \frac{4x \times 3e^{3x} - e^{3x} \times 4}{(4x)^2}$ $= \frac{4e^{3x}(3x - 1)}{16x^2}$ $= \frac{e^{3x}(3x - 1)}{4x^2}$                        | <ul style="list-style-type: none"> <li>• 3 – correct solution</li> <li>• 2 – correct derivative</li> <li>• 1 – attempts to use the quotient rule</li> </ul>      |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                              |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18. | $y = x^{\frac{1}{2}}$ $\frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$ <p>When <math>x = 4</math>,</p> $\frac{dy}{dx} = \frac{1}{2\sqrt{4}} \quad \text{and} \quad y = \sqrt{4}$ $= \frac{1}{4} \quad \quad \quad = 2$ $y - 2 = \frac{1}{4}(x - 4)$ $y = \frac{x}{4} + 1 \quad \text{or} \quad x - 4y + 4 = 0$                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• 3 – correct solution</li> <li>• 2 – finds gradient and y value</li> <li>• 1 – correct derivative</li> </ul>                                                                                                                                         |
| 19. | <p>a)</p>  <p>b)</p> $290^2 = x^2 + 270^2 - 2 \times 270 \times x \times \cos 60^\circ$ $x^2 - 270x - 11200 = 0$ $x = \frac{270 \pm \sqrt{270^2 + 4 \times 11200}}{2}$ $x = 306.5, -36.5$ <p>But <math>x &gt; 0</math>, <math>\therefore</math> The distance is 306.5m.</p>                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• 1 – correct solution</li> <li>• 2 – correct solution</li> <li>• 1 – attempts to use the quadratic formula OR complete the square to solve the equation</li> </ul>                                                                                   |
| 20. | $\frac{6}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}} = \frac{6(\sqrt{5}+\sqrt{3})}{5-3}$ $= 3(\sqrt{5}+\sqrt{3})$                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• 2 – correct solution</li> <li>• 1 – multiplies by conjugate</li> </ul>                                                                                                                                                                              |
| 21. | $\frac{1+\cot \theta}{\operatorname{cosec} \theta} - \frac{\sec \theta}{\tan \theta + \cot \theta} = \sin \theta \left( 1 + \frac{\cos \theta}{\sin \theta} \right) - \frac{1}{\cos \theta \left( \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)}$ $= \sin \theta + \cos \theta - \frac{1}{\sin \theta + \frac{\cos^2 \theta}{\sin \theta}}$ $= \sin \theta + \cos \theta - \frac{\sin \theta}{\sin^2 \theta + \cos^2 \theta}$ $= \sin \theta + \cos \theta - \sin \theta \quad (\text{since } \sin^2 \theta + \cos^2 \theta = 1)$ $= \cos \theta$ | <ul style="list-style-type: none"> <li>• 3 – correct proof</li> <li>• 2 – expresses either fraction with a denominator of 1</li> <li>• 1 – makes some progress using <math>\tan \theta = \frac{\sin \theta}{\cos \theta}</math> or <math>\tan^2 \theta + 1 = \sec^2 \theta</math></li> </ul> |
| 22. | $3^{1-x} = (3^3)^{-\frac{1}{2}}$ $3^{1-x} = 3^{-\frac{3}{2}}$ $1-x = -\frac{3}{2}$ $x = \frac{5}{2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• 2 – correct solution</li> <li>• 1 – expresses <math>\frac{1}{\sqrt{27}}</math> as <math>3^{-\frac{3}{2}}</math></li> </ul>                                                                                                                          |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                        |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23. | $10\sin^2 \theta + \cos \theta - 7 = 0$ $10(1 - \cos^2 \theta) + \cos \theta - 7 = 0$ $10 - 10\cos^2 \theta + \cos \theta - 7 = 0$ $10\cos^2 \theta - \cos \theta - 3 = 0$ $(5\cos \theta - 3)(2\cos \theta + 1) = 0$ $\cos \theta = \frac{3}{5} \quad \text{OR} \quad \cos \theta = \frac{-1}{2}$ $\theta = 0.927..., 5.355... \quad \theta = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$ $\theta = 0.93, 5.36 \quad \theta = \frac{2\pi}{3}, \frac{4\pi}{3}$ $\therefore \theta = 0.93, 2.09, 4.19, 5.36$                                 | <ul style="list-style-type: none"> <li>• 3 - correct solution</li> <li>• 2 - obtains <math>\cos \theta = \frac{3}{5}</math> OR <math>\cos \theta = \frac{-1}{2}</math></li> <li>• 1 - uses <math>\sin^2 \theta + \cos^2 \theta = 1</math> to obtain a quadratic in <math>\cos \theta</math></li> </ul> |
| 24. | $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $= \lim_{h \rightarrow 0} \frac{3(x+h)^2 - 5 - (3x^2 - 5)}{h}$ $= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 5 - 3x^2 + 5}{h}$ $= \lim_{h \rightarrow 0} \frac{6xh + 3h^2}{h}$ $= \lim_{h \rightarrow 0} 6x + 3h$ $= 6x + 0$ $= 6x$                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• 3 - correct solution</li> <li>• 2 - correctly substitutes in to the formula and simplifies the numerator</li> <li>• 1 - correctly substitutes into the formula</li> </ul>                                                                                     |
| 25. | $\Delta ADC,$ $\frac{\sin \angle DAC}{82} = \frac{\sin 26^\circ}{100}$ $\text{In } \sin \angle DAC = \frac{82 \sin 26^\circ}{100}$ $= 0.35946...$ $\angle DAC = 21^\circ 4' \text{ (} 158^\circ 58' \text{ not possible)}$ $\Delta ABC,$ $\text{In } (\theta + \angle DAC) + 90^\circ + 26^\circ = 180^\circ \text{ (Angle sum of triangle)}$ $\theta = 42^\circ 56'$                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• 3 - correct solution</li> <li>• 2 - calculates <math>\angle DAC</math></li> <li>• 1 - writes down correct expression using the sine rule</li> </ul>                                                                                                           |
| 26. | $\cos \angle AOB = \frac{5^2 + 5^2 - (5\sqrt{3})^2}{2 \times 5 \times 5}$ $= \frac{-25}{50}$ $\angle AOB = \frac{2\pi}{3}$ $A = \pi \times 5^2 - \frac{1}{2} \times 5^2 \left( \frac{2\pi}{3} - \sin \frac{2\pi}{3} \right) \text{ OR } A = \frac{1}{2} \times 5^2 \left( \frac{4\pi}{3} - \sin \frac{4\pi}{3} \right)$ $= 25\pi - \frac{25\pi}{3} + \frac{25}{2} \times \frac{\sqrt{3}}{2} \quad = \frac{25}{2} \left( \frac{4\pi}{3} + \frac{\sqrt{3}}{2} \right)$ $= \left( \frac{50\pi}{3} + \frac{25\sqrt{3}}{4} \right) \text{m}^2$ | <ul style="list-style-type: none"> <li>• 4 - correct solution</li> <li>• 3 - correct expression for a segment</li> <li>• 2 - obtains <math>\angle AOB = \frac{2\pi}{3}</math></li> <li>• 1 - correctly substitutes into cosine rule</li> </ul>                                                         |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 27. | <p>a) <math>P(\text{Bag 1 and mint}) = \frac{1}{3} \times \frac{1}{2}</math><br/> <math>= \frac{1}{6}</math></p>                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• 1 – correct answer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|     | <p>b) <math>P(\text{Mint}) = P(\text{Bag 1 and mint}) + P(\text{Bag 2 and mint}) + P(\text{Bag 3 and mint})</math><br/> <math>= \frac{1}{6} + \frac{1}{3} \times \frac{3}{5} + \frac{1}{3} \times \frac{2}{3}</math><br/> <math>= \frac{53}{90}</math></p>                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>• 2 – correct solution</li> <li>• 1 – finds 2 correct cases and adds OR</li> <li>• 1 – finds all 3 cases correctly</li> </ul>                                                                                                                                                                                                                                                                                                                                                |
|     | <p>c) <math>P(\text{Bag 1} \text{mint}) = \frac{P(\text{Bag 1} \cap \text{mint})}{P(\text{mint})}</math><br/> <math>= \frac{\frac{1}{6}}{\frac{53}{90}}</math><br/> <math>= \frac{15}{53}</math></p>                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• 1 – correct answer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 28. |                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• 3 – correct graph</li> <li>• 2 – two of the following features correct: <ul style="list-style-type: none"> <li>- basic shape</li> <li>- x intercept at x = 3 (not tangent)</li> <li>- asymptote at y = 0 (approaching below)</li> </ul> </li> <li>• 1 – one of the following features correct: <ul style="list-style-type: none"> <li>- basic shape</li> <li>- x intercept at x = 3 (not tangent)</li> <li>- asymptote at y = 0 (approaching below)</li> </ul> </li> </ul> |
| 29. | <p><math>N = Ae^{kt}</math><br/> When <math>t = 2, N = 400</math>;<br/> <math>400 = 100e^{2k}</math><br/> <math>e^{2k} = 4</math><br/> <math>2k = \ln 4</math><br/> <math>2k = \ln 2^2</math><br/> <math>2k = 2 \ln 2</math><br/> <math>k = \ln 2</math><br/> <math>\frac{dN}{dt} = k \times 100e^{kt}</math><br/> When <math>t = 10</math>,<br/> <math>\frac{dN}{dt} = \ln 2 \times 100 \times e^{10 \ln 2}</math><br/> <math>= 71000</math> bacteria/minute (nearest 1000)</p> | <ul style="list-style-type: none"> <li>• 4 – Correct solution</li> <li>• 3 – Substitutes correct values of <math>A, k</math> and <math>t</math> into <math>\frac{dN}{dt}</math></li> <li>• 2 – Obtains <math>k = \ln 2</math></li> <li>• 1 – Obtains <math>e^{2k} = 4</math></li> </ul>                                                                                                                                                                                                                             |

| Sample solution |                                                                                                                                                                                                                                                                                         | Suggested marking criteria                                                                                                                                                                                    |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30.             | a) Since the sum of the probability distribution must be 1,<br>$3a + 2b + 0.3 = 1$<br>$3a + 2b = 0.7$                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>• 1 – correct explanation</li> </ul>                                                                                                                                   |
|                 | b) $E(X) = a + 2a + 3a + 4b + 5b + 6 \times 0.3$<br>$4.2 = 6a + 9b + 1.8$<br>$6a + 9b = 2.4$ (1)<br>From (a), $3a + 2b = 0.7$<br>$6a + 4b = 1.4$ (2)<br>(1) – (2) $5b = 1$<br>$b = 0.2$<br>Sub in (1) $3a + 0.4 = 0.7$<br>$3a = 0.3$<br>$a = 0.1$<br>$\therefore a = 0.1$ and $b = 0.2$ | <ul style="list-style-type: none"> <li>• 3 – correct solution</li> <li>• 2 – solves simultaneous equations and finds <math>a</math> or <math>b</math></li> <li>• 1 – equates expected value to 4.2</li> </ul> |
|                 | (c) $Var(X) = E(X^2) - \mu^2$<br>$= 1^2 \times 0.1 + 2^2 \times 0.1 + 3^2 \times 0.1 + 4^2 \times 0.2 + 5^2 \times 0.2 + 6^2 \times 0.3 - (4.2)^2$<br>$= 0.1 + 0.4 + 0.9 + 3.2 + 5 + 10.8 - (4.2)^2$<br>$= 20.4 - 17.64$<br>$= 2.76$                                                    | <ul style="list-style-type: none"> <li>• 2 – correct solution</li> <li>• 1 – finds <math>E(X^2)</math></li> </ul>                                                                                             |